

Exercise Sheet 5: Social Loss, Taylor-Rule Stabilisation, and Optimal Policy

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This sheet accompanies Lectures 7 (Chapter 20) and 8 (Chapters 21–22). The companion starter notebook `week05_new_task_student.ipynb` contains the Python exercises.

Question 1 The Social Loss Function

- (a) Write down the period social loss function used in the lecture. Define each term and explain what the parameter $\kappa > 0$ controls. What does $\kappa \rightarrow \infty$ imply about the preferences of the central bank?
- (b) Derive the iso-loss slope $d\hat{\pi}/d\hat{y}$ along a fixed-loss contour, and find the $\hat{y}$ - and $\hat{\pi}$ -axis intercepts for a contour with loss level v . Sketch one contour for $\kappa = 0.5$ and one for $\kappa = 2$ in the $(\hat{y}, \hat{\pi})$ plane. Describe how the shape changes as κ rises.
- (c) With $\kappa = 2$, compute the social loss at the two points $A = (-1, 1)$ and $B = (0.5, -1.5)$ in $(\hat{y}, \hat{\pi})$ space. Which point lies on the lower iso-loss contour?
- (d) Explain intuitively why the social loss function is also about redistribution. Why is a recession not the same shock for every household?
- (e) (*Python*) Open `week05_new_task_student.ipynb`, Section 2. The code plots iso-loss contours for $\kappa = 0.5$ and $\kappa = 2$ and marks points A and B .
 - i. Verify numerically that A lies on a lower contour than B when $\kappa = 2$.
 - ii. Change κ to 0.5 and recompute the losses at A and B . Does the ranking change? Explain why or why not.

Question 2 The Taylor-Rule AD Curve

- (a) Starting from the IS curve and the Taylor rule

$$y_t - \bar{y} = v_t - \alpha_2(r_t - \bar{r}), \quad i_t = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t,$$

derive the AD curve $\hat{y}_t = z_t - a\hat{\pi}_t$ and express a and z_t in terms of the underlying parameters.

- (b) Complete the table. For each combination of h and b , state the slope of the AD curve and characterise the effect of a positive demand shock $v_t > 0$ on the size of the output shift z_t and on both equilibrium gaps.

h	b	a	AD shape	Demand-shock output shift
0	0			
0	> 0			
> 0	0			
> 0	> 0			

- (c) Suppose $s_t = 0$ and a positive demand shock $v_t > 0$ hits. Explain why policy with $h > 0$ and $b > 0$ can reduce both gaps simultaneously, with no fundamental trade-off.
- (d) Now suppose $z_t = 0$ and a positive supply shock $s_t > 0$ hits. Derive the equilibrium gaps as functions of a and s_t , and explain why stronger policy (larger a) must raise $|\hat{y}|$ whenever it reduces $\hat{\pi}$.
- (e) (*Python*) Open `week05_new_task_student.ipynb`, Section 3. Complete the code to call `equilibrium(par, z, s)` for $h \in \{0, 0.5, 1, 2\}$ and produce a table for a demand shock $z = 2$ and a supply shock $s = 2$.
 - i. For the demand-shock column, verify that both $\hat{y}$ and $\hat{\pi}$ fall as h rises.
 - ii. For the supply-shock column, verify that $\hat{\pi}$ falls but $\hat{y}$ becomes more negative as h rises.
 - iii. The code also plots the trade-off curve in $(\hat{y}, \hat{\pi})$ space as h varies continuously. Describe the shape of this curve and explain why it runs from the upper-right (low h) to the lower-left (high h).

Question 3 Optimal Stabilisation After a Supply Shock

- (a) Write down the one-period optimal policy problem when $\pi_t^e = \pi^*$ and $z_t = 0$. Explain why the origin $(\hat{y}, \hat{\pi}) = (0, 0)$ is infeasible after an adverse supply shock $s_t > 0$.
- (b) Derive the **optimal policy line** from the first-order condition.
- (c) Solve for the optimal responses $\hat{y}^*$ and $\hat{\pi}^*$.
- (d) Use $\gamma = 0.5$, $\kappa = 2$, and $s = 2$. Compute $\hat{y}^*$ and $\hat{\pi}^*$ and verify by substituting into the AS equation.
- (e) A Taylor rule implements the one-period optimum when the AD slope satisfies $a = \gamma\kappa$. With $\alpha_2 = 1.2$ and $b = 0.5$, find the value of h needed to implement the optimum for the parameters in (d).
- (f) (*Python*) Open `week05_new_task_student.ipynb`, Section 4. Complete the code to plot the loss landscape (contours), the shifted AS curve for $s = 2$, the policy line, and the optimal point. Then produce a sensitivity table varying $\kappa \in \{0.25, 0.5, 1, 2, 4, 8\}$.
 - i. Verify that the optimal point lies where AS is tangent to the lowest reachable iso-loss contour.
 - ii. From the sensitivity table, what happens to $\hat{y}^*$ as κ rises? What happens to $\hat{\pi}^*$? Explain the direction of each change.
 - iii. For which value of κ does $a^* = \gamma\kappa$ equal 2? What does this imply for the required h ?

Question 4 Rational Expectations and Credibility

- (a) Define rational expectations in one sentence. Explain why static expectations $\pi_t^e = \pi_{t-1}$ can imply *predictable* forecast errors, and give a concrete example of when this predictability is problematic.

- (b) Write down the impact outcomes for the output gap and inflation gap under the **Policy Ineffectiveness Proposition (PIP)**. State the key information assumption and explain why the Taylor-rule parameters h and b disappear from the impact equations.
- (c) Explain the **inflation-bias problem** in three to four sentences. What institutional arrangement can reduce the bias, and what is the mechanism through which it works?
- (d) (*Python*) Open `week05_new_task_student.ipynb`, Section 5. Complete the code to simulate inflation over $T = 20$ periods after the central bank credibly announces a target reduction from $\pi = 10\%$ to $\pi^* = 2\%$. Under static expectations, the path decays at rate $1/(1 + \gamma a)$ each period. Under credible RE, inflation jumps to π^* immediately.
- Run the simulation with the baseline parameters ($\gamma = 0.5$, $h = 1$, $b = 0.5$, $\alpha_2 = 1.2$). What is the damping factor $\beta = 1/(1 + \gamma a)$?
 - The second panel shows the forecast error under static expectations. What sign does it always have, and why?
 - Suppose the central bank is not believed ($\lambda = 1$, pure static expectations). Compare the social loss in period 3 under static expectations versus credible RE, using $\kappa = 2$. Which regime is less costly?